

CS621/CSL622

Quantum Computing For Computer Scientists

Quantum Architecture

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- Vector spaces
- Linear combination of vectors
- A spanning set of vectors
- Basis and dimension

Reference

- Chapter - Qubits and Quantum States,
- Book - Quantum Computing Explained
- Author - David McMahon

- While the basic unit of information in quantum computation is the qubit, the arena in which quantum computation takes place is a mathematical abstraction called a **vector space**¹.

Definition

A vector space V is a nonempty set with elements u, v called vectors for which the following two operations are defined:

- **Vector addition:** An operation that assigns the sum $w = u + v$, which is also an element of V ; in other words, w is another vector belonging to the same space.
- **Scalar multiplication:** Defines multiplication of a vector by a number α such that the vector $\alpha u \in V$

¹It turns out that quantum states behave mathematically in an analogous way to physical vectors - hence the term vector space

The following axioms hold for the vector space V :

- 1:** Associativity of addition. Given vectors u , v , and w ,

$$(u + v) + w = u + (v + w)$$

- 2:** There is a vector belonging to V called the **zero vector** that satisfies

$$u + 0 = 0 + u = u$$

for any vector $u \in V$

- 3:** For every $u \in V$ there exists an **additive inverse** of u such that

$$u + (-u) = (-u) + u = 0$$

- 4:** Addition of vectors is commutative:

$$u + v = v + u$$

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- Important in quantum computation
- $\mathbb{C}^n \rightarrow$ The vector space of **n-tuples**² of complex numbers
- We label the elements of $\mathbb{C}^n$ by $|a\rangle, |b\rangle, |c\rangle$.
- We write down an element of this vector-space as an n -dimensional column vector or simply a list of numbers $a_1, a_2, \dots, a_n$ arranged in the following way:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

- This type of notation can be used with qubits
- In general, the numbers a_i , which we call the **components** of the vector are complex.

²An ordered collection of numbers

- We write this in a column vector format by putting the coefficient of $|0\rangle$ in the first row and the coefficient of $|1\rangle$ in the second row:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Example

$$|\psi\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} |1\rangle$$

- Hence the column vector representation of this qubit is

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \sqrt{\frac{2}{3}} \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}$$

Multiplication of a vector by a scalar

$$\alpha|a\rangle = \alpha \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \vdots \\ \alpha a_n \end{pmatrix}$$

- This produces another column vector with n complex numbers
- So the result is another element in $\mathbb{C}^n$.
- So $\mathbb{C}^n$ is closed under scalar multiplication.

- Vector addition is carried out component by component, producing a new, third vector:

$$|a\rangle + |b\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}$$

$$|u\rangle = \begin{pmatrix} -1 \\ 7i \\ 2 \end{pmatrix}, \quad |v\rangle = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$

- Compute

$$7|u\rangle + 2|v\rangle$$

Linear Combination

- Let α_i be a set of complex coefficients and $|v_i\rangle$ be a set of vectors.
- A linear combination of these vectors is given by

$$\alpha_1 |v_1\rangle + \alpha_2 |v_2\rangle + \cdots + \alpha_n |v_n\rangle = \sum_{i=1}^n \alpha_i |v_i\rangle$$

- Recall, a **superposition** state of a qubit was also a linear combination of the basic states.

Spanning Set

- If a given set of vectors $|v_1\rangle, |v_2\rangle \cdots |v_n\rangle$ can be used to represent **any** vector $|u\rangle$ that belongs to the **vector space** V , we say that the set $\{|v_i\rangle\}$ **spans** the given vector space.

Example (Three-dimensional vector space $\mathbb{C}^3$)

We can write any 3-D column vector in the following way:

$$|u\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \gamma \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Thus $\{|v_1\rangle, |v_2\rangle, |v_3\rangle\}$ spans $\mathbb{C}^3$ where

$$|v_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |v_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |v_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- Recall that an arbitrary qubit can be written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

- What is the spanning set of a qubit?

Linear Independence

- An important notion involving the linear combination of a set of vectors is that of **linear independence**.
- If $\alpha_1 |v_1\rangle + \alpha_2 |v_2\rangle + \cdots + \alpha_n |v_n\rangle = 0$ and at least one of the $\alpha_i \neq 0$, we say the set $\{|v_i\rangle\}$ is **linearly dependent**
- Alternatively, if *one vector of the set can be written as a linear combination of the other vectors in the set*, then the set is linearly dependent.

Exercise

- Show that the set

$$|a\rangle = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad |b\rangle = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad |c\rangle = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

is linearly dependent; that is, one of the vectors can be written as a linear combination of the other two.

- A spanning set of vectors for a given space V is not unique
- Consider the complex vector space $\mathbb{C}^2$.
- We know that we can write any arbitrary column vector with two elements in the following way

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Thus, the set $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ spans $\mathbb{C}^2$, the vector space in which qubits live.
- However, this is **not the only set** that spans $\mathbb{C}^2$

- Consider the set

$$|u_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |u_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- This set also spans the space $\mathbb{C}^2$
- These vectors can represent qubits.
- We can write any element of $\mathbb{C}^2$ in terms of these vectors

- Consider the arbitrary qubit $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$
- Represent $|\psi\rangle$ in terms of $|u_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |u_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$
- Hint:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \alpha + \alpha \\ \alpha - \alpha \end{pmatrix} + \frac{1}{2} \begin{pmatrix} \beta - \beta \\ \beta + \beta \end{pmatrix}$$

- When a set of vectors is **linearly independent** and they **span** the space, the set is known as a **basis**.
- We can express any vector in the space V in terms of a linear expansion on a basis set.
- The **dimension** of a vector space V is equal to the number of elements in the basis set.
- A vector space can have **many** basis sets.
- Two basis sets for $\mathbb{C}^2$ that we encountered

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad |u_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, |u_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- In term of basis vectors, quantum states in n dimensions are straightforward generalizations of qubits
- A quantum state $|\psi\rangle$ can be written as a linear combination of a basis set $|v_i\rangle$ with complex coefficients of expansion c_i as

$$|\psi\rangle = \sum_{i=1}^n c_i |v_i\rangle = c_1 |v_1\rangle + c_2 |v_2\rangle + \cdots + c_n |v_n\rangle$$

- The modulus squared of a given coefficient c_i gives the probability that **measurement** finds the system in the state $|v_i\rangle$.